$$W = 6.3\times10^{-30}\times3\times10^{5} = 18.9\times10^{-25}\,J$$

For 10^{22} water molecules, the total work done is

$$W_{tot} = 18.9\times10^{-25}\times10^{22} = 18.9\times10^{-3}\,J$$

1.6 GAUSS LAW AND ITS APPLICATIONS

1.6.1 Electric Flux

The number of electric field lines crossing a given area kept normal to the electric field lines is called electric flux. It is usually denoted by the Greek letter Φ_E and its unit is N m^2 C^{-1}. Electric flux is a scalar quantity and it can be positive or negative. For a simpler understanding of electric flux, the following Figure 1.32 is useful.

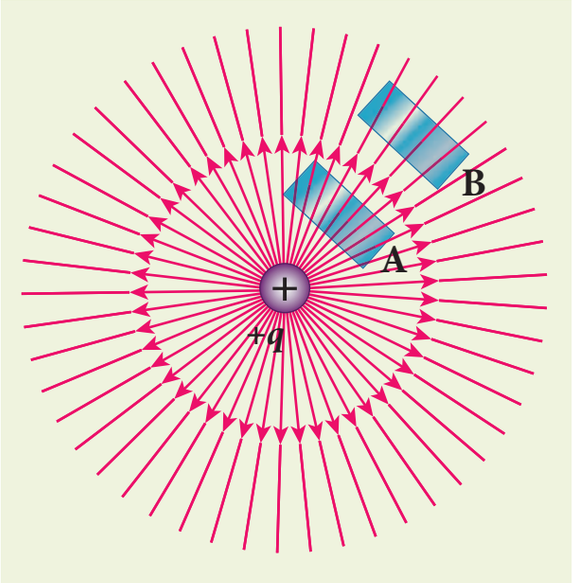


Figure 1.32 Electric flux

The electric field of a point charge is drawn in this figure. Consider two small rectangular area elements placed normal to the field at regions A and B. Even though these elements have the same area, the number of electric field lines crossing the element in region A is more than that crossing the element in region B. Therfore the electric flux in region A is more than that in region B. The electric field strength for a point charge decreases as the distance increases, then for a point charge electric flux also decreases as the distance increases. The above discussion gives a qualitative idea of electric flux. However a precise definition of electric flux is needed.

Electric flux for uniform Electric field

Consider a uniform electric field in a region of space. Let us choose an area A normal to the electric field lines as shown in Figure 1.33 (a). The electric flux for this case is

$$\Phi_E = EA \qquad (1.52)$$

Suppose the same area A is kept parallel to the uniform electric field, then no electric field lines pierce through the area A , as shown in Figure 1.33(b). The electric flux for this case is zero.

$$\Phi_E = 0 \qquad (1.53)$$

If the area is inclined at an angle θ with the field, then the component of the electric field perpendicular to the area alone contributes to the electric flux. The electric field component parallel to the surface area will not contribute to the electric flux. This is shown in Figure 1.33 (c). For this case, the electric flux

$$\Phi_E = (E\cos\theta)\,A \qquad (1.54)$$

Further, θ is also the angle between the electric field and the direction normal to the area. Hence in general, for uniform electric field, the electric flux is defined as